



Advanced Time Series Analysis and Forecasting

Seminar 15: Review and project defence



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Acronyms used in this material

- ▣ **ACI**: Adaptive Conformal Inference
- ▣ **AI**: Artificial Intelligence
- ▣ **AR**: AutoRegressive (model); in event studies: Abnormal Return
- ▣ **BET**: Bucharest Exchange Trading (index)
- ▣ **BNR**: Banca Națională a României (the National Bank of Romania)
- ▣ **CW**: Clark–West (test for nested models)
- ▣ **DM**: Diebold–Mariano (test)
- ▣ **EODHD**: EOD Historical Data (market-data provider)
- ▣ **FRED**: Federal Reserve Economic Data
- ▣ **FWER**: Family-Wise Error Rate
- ▣ **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GDP**: Gross Domestic Product
- ▣ **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ **HLN**: Harvey–Leybourne–Newbold (small-sample correction of the DM test)
- ▣ **LR**: Likelihood Ratio (test)
- ▣ **MA**: Moving Average
- ▣ **QML**: Quasi-Maximum Likelihood
- ▣ **RMSE**: Root Mean Squared Error
- ▣ **RMSPE**: Root Mean Squared Prediction Error
- ▣ **SVAR**: Structural Vector AutoRegression
- ▣ **US**: United States
- ▣ **VAR**: Vector AutoRegression
- ▣ **VECM**: Vector Error Correction Model

Today's question and route

- **Question:** can you use the tools of the whole course quickly and defend a replication result under questioning?
 - ▶ this seminar comes **before** Lecture 15 and is a project clinic: the section “Prerequisites for Today” is a formula sheet of the course
- **Route**
 - ▶ Part A: ten short problems from [Chapters 0–14](#), of the kind asked at the defence
 - ▶ Part B: four mock defences on replication outputs (Meese–Rogoff for EUR/RON, the term spread, a leakage audit, a rule search on the BET)
 - ▶ Part C: the pre-registration of your own project; an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

Exercise map

Task	Question	Type	Model
A1, A2	the long-run variance of a mean (Ch. 0); a DM test by hand (Ch. 1)	Solved, Proposed	A1
A3, A4	Cholesky responses (Ch. 3); the half-life of an equilibrium error (Ch. 4)	Solved, Proposed	A3
A5, A6	one Kalman step (Ch. 6); a two-state Markov chain (Ch. 7)	Solved, Proposed	A5
A7, A8	the Kupiec test (Ch. 9); GARCH persistence (Ch. 8)	Solved, Proposed	A7
A9, A10	the split-conformal quantile (Ch. 13); placebo p-values and Holm (Ch. 14, 0)	Solved, Proposed	A9
B1, B2	mock defences: Meese–Rogoff for EUR/RON; the term spread	Solved, Proposed	B1
B3, B4	mock defences: a leakage audit; a rule search on the BET	Solved, Proposed	B3
C1, C2	your pre-registration with a power calculation; what is wrong in an AI answer?	Proposed	B1, A1–A10

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
EUR/RON reference rate	BNR	monthly (end of month)	2005–2026
US real GDP, 10-year and 3-month Treasury yields	FRED	quarterly	1962–2025
BET index	EODHD	daily	2001–2012
Simulated data (leakage audit)	–	–	–

- In the notebook: `dm_test`, `lr_v`, `ols_se`, `leakage_r2`, `stationary_bootstrap_index`, `p_required`; no account or key is needed

Prerequisites for Today (1/20): the long-run variance

- ▣ The **long-run variance** sets the precision of a mean of dependent data: $\text{Var}(\bar{x}) \approx \Omega/T$

$$\Omega = \gamma_0 + 2 \sum_{j \geq 1} \gamma_j$$

- ▶ γ_j : autocovariance at lag j (γ_0 : the variance); T : sample size

- ▣ **AR(1)** $x_t = \phi x_{t-1} + \varepsilon_t$, innovation variance σ^2

$$\gamma_0 = \frac{\sigma^2}{1 - \phi^2}, \quad \Omega = \frac{\sigma^2}{(1 - \phi)^2}, \quad \frac{\Omega}{\gamma_0} = \frac{1 + \phi}{1 - \phi}$$

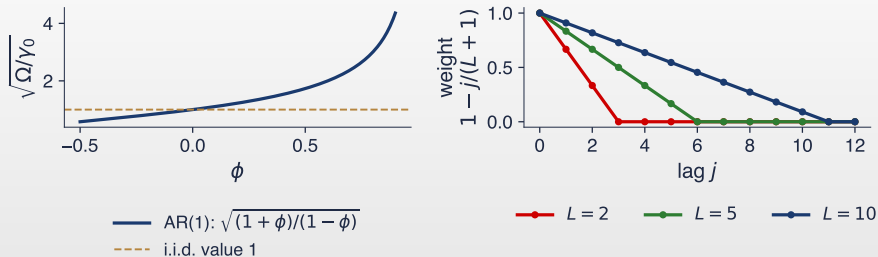
- ▶ $\sqrt{\Omega/\gamma_0}$: the factor by which i.i.d. standard errors are too small

- ▣ **Newey–West** estimator [Newey and West \(1987\)](#): weighted sum of sample autocovariances

$$\hat{\Omega} = \hat{\gamma}_0 + 2 \sum_{j=1}^L \left(1 - \frac{j}{L+1}\right) \hat{\gamma}_j, \quad \hat{\gamma}_j = \frac{1}{T} \sum_{t>j} (x_t - \bar{x})(x_{t-j} - \bar{x})$$

- ▶ L : number of lags (rule $L = \lfloor 4(T/100)^{2/9} \rfloor$, $\lfloor \cdot \rfloor$: integer part); $1 - j/(L+1)$: Bartlett weights

Prerequisites for Today (2/20): dependence and the precision of a mean



- Left: the factor $\sqrt{\Omega/\gamma_0}$ by which i.i.d. standard errors must be multiplied for an $\text{AR}(1)$
 - ▶ 1.36 for $\phi = 0.3$, 4.36 for $\phi = 0.9$; below 1 for $\phi < 0$ (negative autocorrelation)
- Right: the Bartlett weights $1 - j/(L+1)$ of $\hat{\gamma}_j$ in $\hat{\Omega}$; they decline linearly and vanish after lag L
 - ▶ the rule $L = \lfloor 4(T/100)^{2/9} \rfloor$ gives $L = 4$ for $T = 250$ and $L = 6$ for $T = 1000$; the weights keep $\hat{\Omega} \geq 0$

Prerequisites for Today (3/20): comparing two forecasts

- **DM test** Diebold and Mariano (1995): is the expected loss of forecast 1 equal to that of forecast 2?

$$d_t = L(e_{1t}) - L(e_{2t}), \quad \text{DM} = \bar{d} / \sqrt{\hat{\Omega}_d / P}$$

- ▶ e_{it} : forecast error of model i ; L : loss (e.g. squared error); $\bar{d}$: mean of d_t ; $\hat{\Omega}_d$: its long-run variance; P : number of forecasts

- **HLN correction** Harvey, Leybourne and Newbold (1997) for small samples and horizon h

$$\text{DM}^* = \text{DM} \sqrt{(P + 1 - 2h + h(h - 1)/P) / P}$$

- ▶ compared with Student- t with $P - 1$ degrees of freedom (t_{P-1})

- **Clark–West** Clark and West (2007) for nested models (model 1 is the smaller one)

$$f_t = e_{1t}^2 - [e_{2t}^2 - (\hat{y}_{1t} - \hat{y}_{2t})^2]$$

- ▶ $\hat{y}_{it}$: forecast of model i ; the bracket removes the noise of estimating the extra coefficients; one-sided test of $\mathbb{E}f_t \leq 0$

Prerequisites for Today (4/20): accuracy measures and nested models

- **RMSE** of P forecasts and the ratio to a benchmark

$$\text{RMSE} = \sqrt{\frac{1}{P} \sum_{t=1}^P e_t^2}, \quad \text{ratio} = \frac{\text{RMSE}_{\text{model}}}{\text{RMSE}_{\text{benchmark}}}$$

- ▶ ratio < 1: the model is more accurate; random walk for a log change: $\hat{y}_{t+1} = 0$

- **Clark–West** Clark and West (2007) statistic: the mean of the adjusted differential f_t over its standard error

$$\text{CW} = \frac{\bar{f}}{\sqrt{\hat{\Omega}_f/P}}, \quad \bar{f} = \frac{1}{P} \sum_{t=1}^P f_t$$

- ▶ $\hat{\Omega}_f$: the long-run variance of f_t ; reject the smaller model if $\text{CW} > 1.645$ (one-sided, 5%)

- **Out-of-sample** R^2 : positive when the forecast beats the training mean $\bar{y}_{\text{train}}$

$$R_{\text{OOS}}^2 = 1 - \frac{\sum_t (y_t - \hat{y}_t)^2}{\sum_t (y_t - \bar{y}_{\text{train}})^2}$$

- **t-statistic** of a slope: $t = \hat{\beta}/\text{se}(\hat{\beta})$, with a classical or a HAC standard error (Newey–West Newey and West (1987), L lags)

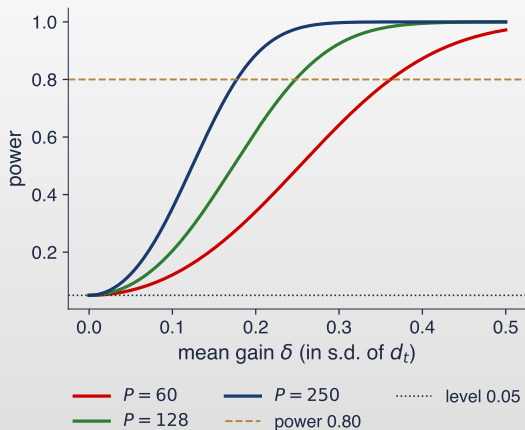
Prerequisites for Today (5/20): the power of a DM test

- ▣ Number of forecasts P needed to detect a mean gain δ with power π

$$P \approx \Omega_d \frac{(z_{1-a/2} + z_\pi)^2}{\delta^2}$$

- ▶ δ : the mean gain, in standard deviations of d_t ; a : the test level (e.g. 5%); z_q : the q -quantile of the standard Normal ($z_{0.975} = 1.96$, $z_{0.8} = 0.84$)
- ▶ Ω_d : long-run variance of the standardised d_t ; for an AR(1) differential with coefficient ρ , $\Omega_d = (1 + \rho)/(1 - \rho)$
- ▣ Reading
 - ▶ halving δ quadruples P ; positive dependence ($\rho > 0$) multiplies P by Ω_d
 - ▶ a non-rejection with P far below this value says little about the absence of a gain

Prerequisites for Today (6/20): power curves of the DM test



- Power of a two-sided 5% test, i.i.d. differential ($\Omega_d = 1$)

$$\pi(\delta) = \Phi(\delta\sqrt{P} - 1.96) + \Phi(-\delta\sqrt{P} - 1.96)$$

- ▶ Φ : the standard Normal distribution function; δ : the mean gain in standard deviations of d_t

- For $\delta = 0.2$: power 0.34 with $P = 60$, 0.62 with $P = 128$, 0.89 with $P = 250$

- ▶ at $\delta = 0$ every curve equals the level 0.05

- With dependent d_t , replace $\sqrt{P}$ by $\sqrt{P/\Omega_d}$: the curves move to the right

Prerequisites for Today (7/20): structural VAR and cointegration

- ▣ **Recursive SVAR**: the reduced-form errors are a lower-triangular combination of structural shocks

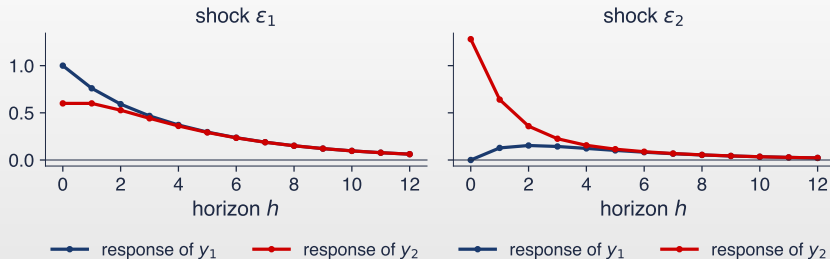
$$u_t = B_0 \varepsilon_t, \quad B_0 = \text{chol}(\Sigma_u), \quad \Theta_h = A^h B_0 \text{ (VAR(1))}$$

- ▶ Σ_u : covariance of u_t ; chol: Cholesky factor; A : VAR(1) coefficient matrix; Θ_h : responses at horizon h
- ▶ 2×2 : $b_{11} = \sqrt{\sigma_{11}}$, $b_{21} = \sigma_{21}/b_{11}$, $b_{22} = \sqrt{\sigma_{22} - b_{21}^2}$, with σ_{ij} the elements of Σ_u

- ▣ **VECM** without short-run terms: $\Delta y_t = \alpha \beta' y_{t-1} + \varepsilon_t$

- ▶ β : cointegrating vector; α : adjustment speeds; $z_t = \beta' y_t$: the equilibrium error, which follows an AR(1)
- ▶ $z_t = (1 + \beta' \alpha) z_{t-1} + \beta' \varepsilon_t$; half-life of a deviation: $\ln 0.5 / \ln(1 + \beta' \alpha)$ periods

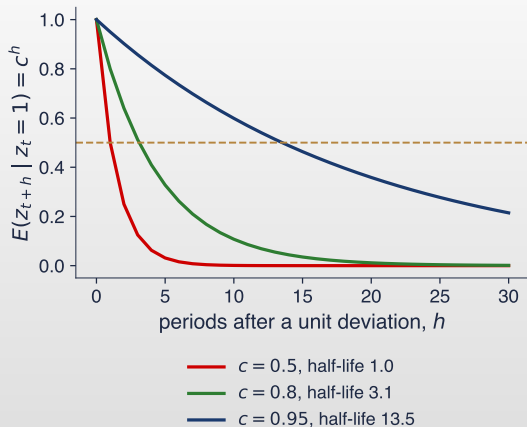
Prerequisites for Today (8/20): Cholesky impulse responses



□ Illustration: $A = \begin{pmatrix} 0.7 & 0.1 \\ 0.3 & 0.5 \end{pmatrix}$, $\Sigma_u = \begin{pmatrix} 1 & 0.6 \\ 0.6 & 2 \end{pmatrix}$; responses $\Theta_h = A^h B_0$ for $h = 0, \dots, 12$

- ▶ B_0 : $b_{11} = 1$, $b_{21} = 0.60$, $b_{22} = 1.28$; $b_{12} = 0$, so y_1 does not respond on impact to ε_2 (right, $h = 0$)
- ▶ element (i, k) of Θ_h : the response of y_i after h periods to a one-standard-deviation shock ε_k ; stable VAR: all responses die out

Prerequisites for Today (9/20): the half-life of a deviation



- An AR(1) deviation $z_t = cz_{t-1} + \varepsilon_t$, $0 < c < 1$, decays on average as c^h

$$c^{h^*} = 0.5 \iff h^* = \frac{\ln 0.5}{\ln c}$$

- ▶ h^* : the half-life, in periods; here 1.0, 3.1 and 13.5 for $c = 0.5, 0.8, 0.95$
- The same formula serves the equilibrium error of a VECM ($c = 1 + \beta'\alpha$) and a GARCH volatility shock ($c = \alpha + \beta$)
- ▶ c close to 1: very slow return to equilibrium

Prerequisites for Today (10/20): the Kalman filter for a local level

- ▣ **Local level:** the observation is a random-walk level plus noise

$$y_t = \mu_t + \varepsilon_t, \quad \mu_{t+1} = \mu_t + \eta_t$$

- ▶ $\sigma_\varepsilon^2, \sigma_\eta^2$: variances of the observation and level noises; a, P : predicted level and its variance

- ▣ **One step:** prediction error variance, gain, update, next prediction

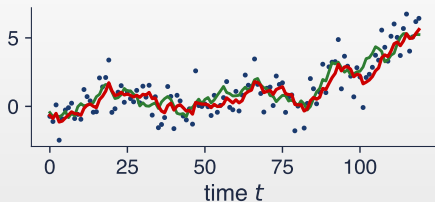
$$F = P + \sigma_\varepsilon^2, \quad K = P/F, \quad a_{t|t} = a + K(y - a), \quad P_{t|t} = P(1 - K), \quad P_{t+1} = P_{t|t} + \sigma_\eta^2$$

- ▶ $K \in (0, 1)$: the share of the surprise $y - a$ added to the level

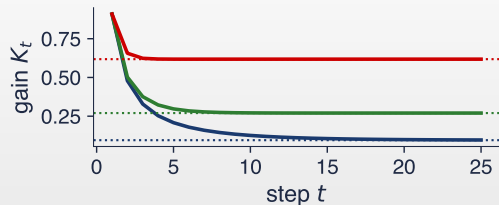
- ▣ **Steady state:** depends only on the signal-to-noise ratio $q = \sigma_\eta^2 / \sigma_\varepsilon^2$

$$\bar{P} = \sigma_\varepsilon^2 \frac{q + \sqrt{q^2 + 4q}}{2}$$

Prerequisites for Today (11/20): the Kalman filter at work



• y_t — level μ_t — filtered $a_{t|t}$



— $q = 0.01$ — $q = 0.1$ — $q = 1$

- Left: a simulated local level with $\sigma_\epsilon^2 = 1$, $\sigma_\eta^2 = 0.1$; the filtered level follows the true level and smooths the noise
- Right: the gain K_t from a vague start ($P_1 = 10$) converges in a few steps to $\bar{K} = \bar{P}/(\bar{P} + \sigma_\epsilon^2)$ (dotted)
 - ▶ $\bar{K} = 0.095, 0.270, 0.618$ for $q = 0.01, 0.1, 1$: a noisier level (larger q) gives more weight to the new observation
 - ▶ with K constant the filter is exponential smoothing: $a_{t|t} = (1 - \bar{K})a_{t-1|t-1} + \bar{K}y_t$

Prerequisites for Today (12/20): Markov chains and GARCH

- ▣ **Two-state Markov chain:** p_{ij} is the probability of staying in regime i from one period to the next

$$\mathbb{E}D_i = \frac{1}{1 - p_{ii}}, \quad \pi_1 = \frac{1 - p_{22}}{2 - p_{11} - p_{22}}, \quad \lambda = p_{11} + p_{22} - 1$$

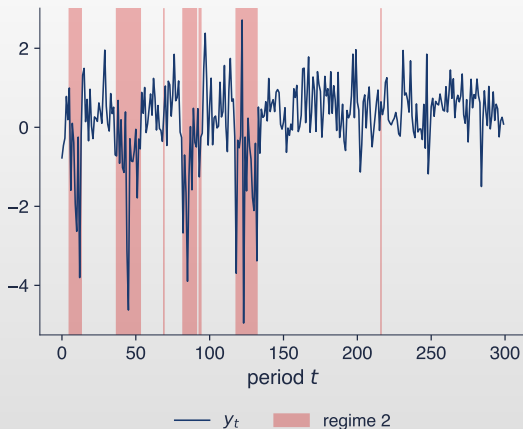
- ▶ $\mathbb{E}D_i$: expected duration of regime i ; π_1 : long-run (ergodic) share of time in regime 1; λ : persistence of the chain

- ▣ **GARCH(1,1):** today's variance from yesterday's squared shock and variance

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad \sigma^2 = \frac{\omega}{1 - \alpha - \beta}$$

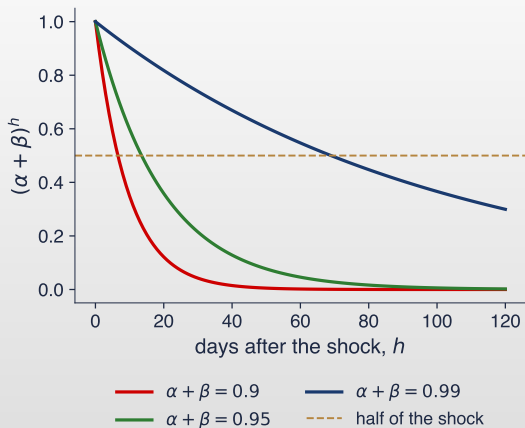
- ▶ $\omega > 0, \alpha, \beta \geq 0$; ε_{t-1} : yesterday's shock; σ^2 : unconditional variance (when $\alpha + \beta < 1$)
- ▶ persistence $\alpha + \beta$; half-life of a volatility shock $\ln 0.5 / \ln(\alpha + \beta)$ periods

Prerequisites for Today (13/20): a simulated two-state Markov chain



- $p_{11} = 0.97$ (calm regime), $p_{22} = 0.90$ (turbulent regime, shaded); y_t has a lower mean and a larger variance in regime 2
- Expected durations 33.3 and 10.0 periods; ergodic share of regime 1 $\pi_1 = 0.769$
 - ▶ in this sample of 300 periods regime 1 occupies 0.817 of the time
 - ▶ $\lambda = 0.87$: the regime of today predicts the regime of the next periods
- The duration of a regime is geometric:
 $\Pr(D_i = d) = p_{ii}^{d-1}(1 - p_{ii})$, $d = 1, 2, \dots$

Prerequisites for Today (14/20): the decay of a volatility shock



- GARCH(1,1) variance forecast h days ahead

$$\mathbb{E}_t \sigma_{t+h}^2 - \sigma^2 = (\alpha + \beta)^{h-1} (\sigma_{t+1}^2 - \sigma^2)$$

- ▶ σ^2 : the unconditional variance; a deviation of today's variance from σ^2 shrinks by the factor $\alpha + \beta$ each day
- Half-lives: 6.6, 13.5 and 69.0 days for $\alpha + \beta = 0.90$, 0.95, 0.99
- ▶ near 1 the half-life explodes: small changes in the persistence change it a lot

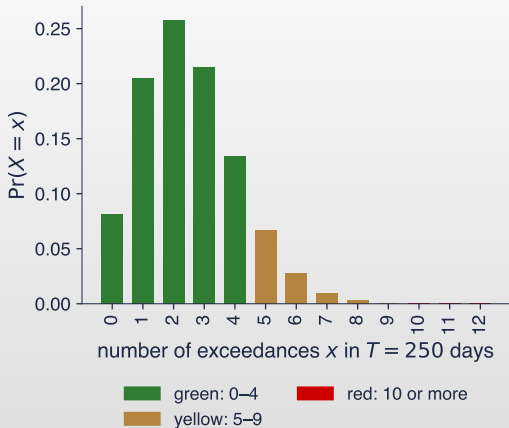
Prerequisites for Today (15/20): VaR backtests

- ▣ VaR 1%: $\text{VaR}_\alpha = -q_\alpha$, with q_α the α -quantile of the return and $\alpha = 0.01$
- ▣ **Kupiec test**: x exceedances in T days; is the exceedance rate equal to α ?

$$LR_{uc} = -2 \ln \frac{(1 - \alpha)^{T-x} \alpha^x}{(1 - \hat{\pi})^{T-x} \hat{\pi}^x} \sim \chi_1^2, \quad \hat{\pi} = x/T$$

- ▶ numerator: likelihood of the exceedances under the target rate α ; denominator: under the observed rate $\hat{\pi}$
- ▶ a large LR_{uc} (small p-value): too many or too few exceedances
- ▣ **Basel traffic light** for 250 days at 1%
 - ▶ green 0–4, yellow 5–9, red 10 or more exceedances

Prerequisites for Today (16/20): the binomial law of exceedances



- If the VaR is correct, the number of exceedances X in T days is Binomial(T, α)

$$\Pr(X = x) = \binom{T}{x} \alpha^x (1 - \alpha)^{T-x}$$

- ▶ $\binom{T}{x}$: the number of ways to place x exceedances among T days; $\mathbb{E}X = T\alpha = 2.5$ for $T = 250$, $\alpha = 0.01$
- The exact one-sided p-value of x_0 exceedances is the tail $\Pr(X \geq x_0) = \sum_{x=x_0}^T \Pr(X = x)$
- Basel zones for a correct model: green with probability 89.2%, yellow 10.8%, red 0.03%
 - ▶ a correct model falls in the yellow zone about one year in ten

Prerequisites for Today (17/20): conformal prediction

- ▣ **Split conformal** Romano, Patterson and Candès (2019): scores $S_1, \dots, S_n$ on a calibration set, e.g. absolute residuals

$$\hat{q} = S_{(k)}, \quad k = \lceil (n+1)(1-\alpha) \rceil$$

- ▶ $S_{(k)}$: the k -th smallest score; $\lceil \cdot \rceil$: rounding up; α : the allowed miss rate; the interval is $\hat{y} \pm \hat{q}$

- ▣ **Guarantee** under exchangeability (no ties)

$$1 - \alpha \leq \Pr(Y_{n+1} \in \hat{C}) \leq 1 - \alpha + \frac{1}{n+1}$$

- ▶ $\hat{C}$: the prediction interval; exchangeable: the joint law does not change when the points are reordered

- ▣ **ACI** Gibbs and Candès (2021) for dependent data: the working level α_t is updated after each observation

- ▶ $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$; γ : step; $\text{err}_t = 1$ after a miss: a miss widens the next interval

Prerequisites for Today (18/20): searches, placebos and the defence

- ▣ **Multiple tests:** m tests at level α , p-values p_j , sorted $p_{(1)} \leq \dots \leq p_{(m)}$
 - ▶ Bonferroni: reject when $p_j \leq \alpha/m$
 - ▶ Holm: reject while $p_{(i)} \leq \alpha/(m - i + 1)$, stop at the first failure; both control the FWER (probability of at least one false rejection)
 - ▶ max- t (Reality Check logic [White \(2000\)](#)): compare the best t with the bootstrap distribution of $\max_k t_k^*$, resampled jointly in blocks
- ▣ **Synthetic control placebo** [Abadie, Diamond and Hainmueller \(2015\)](#): rank of the treated unit among $J + 1$ units
 - ▶ p-value = $\#\{j : r_j \geq r_1\}/(J + 1)$; r_j : post/pre RMSPE ratio of unit j ; the smallest p-value is $1/(J + 1)$
- ▣ **Leakage:** any choice (selection, scaling, tuning) that uses data after the forecast origin [Kaufman et al. \(2012\)](#)
- ▣ At the defence: individual, three blocks of questions (code, results, chapters), no AI tools
 - ▶ a good answer: the number, its uncertainty, the assumption behind it, what would change it

Prerequisites for Today (19/20): a rule search in formulas

- ▣ **Moving-average rule** with window n : long the next day if the close is above its n -day average, otherwise cash

$$s_{n,t} = 1 \left\{ P_t > \frac{1}{n} \sum_{i=0}^{n-1} P_{t-i} \right\}, \quad f_{n,t+1} = r_{t+1}(s_{n,t} - 1)$$

- ▶ P_t : the close; r_{t+1} : the next log return; $f_{n,t+1}$: the return of the rule minus that of buy-and-hold
- ▶ $t_n = \bar{f}_n / \text{se}_{\text{HAC}}(\bar{f}_n)$ for each of the K rules; the best rule has $t_{\max} = \max_n t_n$

- ▣ **Stationary bootstrap**: resample the days in blocks of random length, geometric with mean b , wrapping around the end

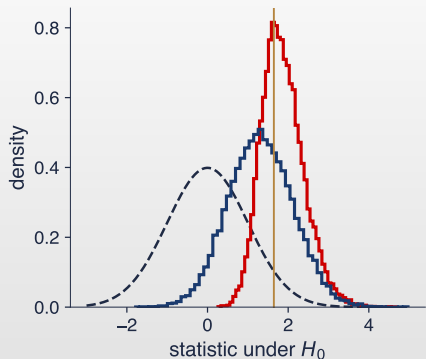
- ▶ the same days for all K rules: the correlation of the rules is kept; blocks keep the dependence over time

- ▣ **Max- t p-value** with B resamples of the centred differentials $f_{n,t} - \bar{f}_n$

$$t_{\max}^{*(b)} = \max_n \frac{\bar{f}_n^{*(b)} - \bar{f}_n}{\text{se}(\bar{f}_n)}, \quad p = \frac{1}{B} \sum_{b=1}^B 1 \{ t_{\max}^{*(b)} \geq t_{\max} \}$$

- ▶ the critical value is the 95% quantile of $t_{\max}^*$, not 1.645

Prerequisites for Today (20/20): the maximum of many t -statistics



- one t , $N(0, 1)$
- max of 20, correlation 0.5
- max of 20, correlation 0
- 1.645 (one test, 5%)

- Simulated: 20 statistics $N(0, 1)$ under H_0 , independent or with pairwise correlation 0.5
 - ▶ the maximum is far to the right of a single $N(0, 1)$
- 95% quantiles of the maximum: 2.79 (independent), 2.65 (correlation 0.5), against 1.645 for one test
 - ▶ with 20 independent tests at 5%, at least one rejection occurs with probability $1 - 0.95^{20} = 64\%$
 - ▶ the more correlated the rules, the closer the maximum stays to a single test

A1: how much does a mean know? (Chapter 0) [Solved]

Task

- A series is AR(1) with $\phi = 0.6$ and $\sigma^2 = 1$. A team reports $t = 2.5$ for its mean with i.i.d. standard errors.
- 1. Compute γ_0 and Ω .
- 2. By what factor is the i.i.d. standard error too small?
- 3. Compute the corrected t and state the conclusion at 5%.
- Report: two numbers; one number; one number and one sentence.

Solution

- 1. $\gamma_0 = 1/(1 - 0.36) = 1.5625$; $\Omega = 1/(1 - 0.6)^2 = 6.25$
- 2. $\Omega/\gamma_0 = (1 + \phi)/(1 - \phi) = 4$: the standard error is too small by a factor of $\sqrt{4} = 2$
- 3. $t = 2.5/2 = 1.25 < 1.96$: not significant
- Interpretation: at the defence, the first question about any t -statistic is "which standard error?"

A2: a Diebold–Mariano test by hand (Chapter 1) [Proposed]

Solution discussed in class

Task

- ▣ Squared-error differentials d_t (benchmark minus model), $h = 1$: 0.8, -0.2, 0.5, 1.1, 0.3, -0.4, 0.9, 0.6, 0.2, 0.4.
Model: A1.
- ▣ 1. Compute $\bar{d}$, $\hat{\gamma}_0$ and $\hat{\gamma}_1$ (divide by P).
- ▣ 2. Compute $\hat{\Omega}$ with the Bartlett kernel and $L = 1$, then DM and HLN.
- ▣ 3. Compare with the critical value of t_9 and state the conclusion.
- ▣ Report: three numbers; three numbers; one sentence.

A3: Cholesky responses (Chapter 3) [Solved]

Task

- A VAR(1) $y_t = Ay_{t-1} + u_t$ with $A = \begin{pmatrix} 0.5 & 0 \\ 0.2 & 0.4 \end{pmatrix}$,
 $\Sigma_u = \begin{pmatrix} 4 & 2 \\ 2 & 5 \end{pmatrix}$, recursive ordering.
- 1. Compute B_0 .
- 2. Compute $\Theta_1 = AB_0$ and $\Theta_2 = A^2B_0$.
- 3. What changes if the ordering is reversed?
- Report: a matrix; two matrices; one sentence.

Solution

- 1. $b_{11} = \sqrt{4} = 2.0$, $b_{21} = 2/2 = 1.0$, $b_{22} = \sqrt{5-1} = 2.0$
- 2. $\Theta_1 = \begin{pmatrix} 1.00 & 0.00 \\ 0.80 & 0.80 \end{pmatrix}$, $\Theta_2 = \begin{pmatrix} 0.50 & 0.00 \\ 0.52 & 0.32 \end{pmatrix}$
- 3. Reversed, the first variable responds on impact to the second shock and the second no longer to the first; with $\sigma_{12} \neq 0$ all responses change: the ordering is the identifying assumption
- Interpretation: a defence question is always "why this ordering?"

A4: the half-life of an equilibrium error (Chapter 4) [Proposed]

Solution discussed in class

Task

- A bivariate VECM with $\beta = (1, -1)'$ and $\alpha = (-0.2, 0.05)'$, no short-run terms. Model: A3.
- 1. Derive the AR(1) of $z_t = \beta' y_t$ and its coefficient.
- 2. Compute the half-life of a deviation.
- 3. Which variable is weakly exogenous if $\alpha_2 = 0$, and how is that tested?
- Report: one equation and one number; one number; one sentence.

A5: one step of the Kalman filter (Chapter 6) [Solved]

Task

- Local level with $\sigma_\varepsilon^2 = 4$, $\sigma_\eta^2 = 1$; predicted state $a = 10$, $P = 2$; observation $y = 13$.
- 1. Compute v , F and K .
- 2. Compute $a_{t|t}$, $P_{t|t}$ and P_{t+1} .
- 3. Compute the steady-state $\bar{P}$ and gain.
- Report: three numbers; three numbers; two numbers.

Solution

- 1. $v = 3$, $F = 2 + 4 = 6$, $K = 2/6 = 0.333$
- 2. $a_{t|t} = 10 + K \cdot 3 = 11$; $P_{t|t} = 2(1 - K) = 1.333$; $P_{t+1} = 2.333$
- 3. $q = 0.25$: $\bar{P} = 2.562$, $\bar{K} = \bar{P}/(\bar{P} + 4) = 0.390$
- Interpretation: the filter moves one third of the way towards the surprise; the signal-to-noise ratio q alone sets the long-run gain

A6: a two-state Markov chain (Chapter 7) [Proposed]

Solution discussed in class

Task

- ▣ Regime 1 (expansion) with $p_{11} = 0.95$, regime 2 (recession) with $p_{22} = 0.80$, monthly data. Model: A5.
- ▣ 1. Compute the expected durations.
- ▣ 2. Compute the ergodic probability of regime 1 and the persistence λ .
- ▣ 3. Why can a likelihood-ratio test of one regime against two not use χ^2_2 ?
- ▣ Report: two numbers; two numbers; one sentence.

A7: the Kupiec test for VaR 1% (Chapter 9) [Solved]

Task

- A VaR 1% model has $x = 6$ exceedances in $T = 250$ days.
- 1. Compute the hit rate and the expected number of exceedances.
- 2. Compute LR_{uc} and its p-value; compare with the exact binomial tail $\Pr(X \geq 6)$.
- 3. Give the Basel zone and the conclusion.
- Report: two numbers; three numbers; one sentence.

Solution

- 1. 2.4%; expected 2.5
- 2. $LR_{uc} = 3.56$, $p = 0.059$; exact one-sided binomial $p = 0.041$
- 3. Yellow zone; the asymptotic test does not reject at 5%, the exact tail does: with $\alpha = 1\%$ and 250 days, report both
- Interpretation: backtests on a few tail days have little power; one exceedance changes the verdict

A8: GARCH persistence (Chapter 8) [Proposed]

Solution discussed in class

Task

- ▣ Daily returns in %, GARCH(1,1) with $\omega = 0.02$, $\alpha = 0.08$, $\beta = 0.90$. Model: A7.
- ▣ 1. Compute the persistence and the half-life of a volatility shock.
- ▣ 2. Compute the unconditional variance and the annualised volatility (250 days).
- ▣ 3. Why are Hessian standard errors of α misleading with fat-tailed returns?
- ▣ Report: two numbers; two numbers; one sentence.

A9: the split-conformal quantile (Chapter 13) [Solved]

Task

- ▣ Absolute residuals of 19 calibration points: 0.12, 0.85, 0.40, 1.30, 0.07, 0.66, 2.10, 0.95, 0.31, 1.75, 0.52, 0.22, 1.05, 0.48, 0.90, 1.42, 0.18, 0.73, 1.20; $\alpha = 0.1$.
- ▣ 1. Compute k and $\hat{q}$.
- ▣ 2. Give the coverage bounds of $\hat{y} \pm \hat{q}$.
- ▣ 3. Which assumption fails for daily returns, and what replaces it?
- ▣ Report: two numbers; two numbers; one sentence.

Solution

- ▣ 1. $k = \lceil 20 \cdot 0.9 \rceil = 18$: the 18th smallest score, $\hat{q} = 1.75$
- ▣ 2. Between 0.90 and $0.9 + 1/20 = 0.950$, marginally
- ▣ 3. Exchangeability fails with volatility clustering; ACI or weighted conformal restore long-run coverage, conditional coverage must be tested
- ▣ Interpretation: a guarantee is only as good as its assumption; say which one you rely on

A10: placebo p-values and Holm (Chapters 14 and 0) [Proposed]

Solution discussed in class

Task

- ▣ A synthetic control with 19 donors; the treated unit has the largest post/pre RMSPE ratio. The team tests five outcomes with $p = 0.010, 0.020, 0.040, 0.200, 0.030$. Model: A9.
- ▣ 1. Compute the placebo p-value and the smallest attainable value.
- ▣ 2. How many outcomes are significant at 5% without correction, with Bonferroni and with Holm?
- ▣ 3. What should the pre-registration have said?
- ▣ Report: two numbers; three counts; one sentence.

B1: Meese and Rogoff for EUR/RON [Solved]

- **Question:** a team replicates [Meese and Rogoff \(1983\)](#): can an AR(1) beat the random walk for the monthly EUR/RON change? You are the examiner and the candidate at once
- **Inputs:** BNR reference rate, end of month, $100 \times \log$ change; rolling window of 60 months; 128 one-month forecasts from January 2016
- **Tasks**
 1. Produce the output: RMSE of both models, their ratio, DM–HLN and Clark–West.
 2. Defence question: why is Clark–West the right test here, and what does it test?
 3. Defence question: is the non-rejection evidence that the random walk is best? Use the power formula.
 4. Interpretation: write the one-sentence conclusion you would defend.
- **Report:** four numbers and two test statistics, two short answers, one sentence
- **Notebook:** section B1

B1: solution [Solved]



- RMSE: random walk 0.509, AR(1) 0.524; ratio 1.028; DM–HLN = 1.19 ($p = 0.235$); Clark–West = 1.11 (one-sided $p = 0.134$)
- Clark–West: the random walk is nested in the AR(1); under the null the estimated coefficients add noise to the larger model, which biases DM towards the random walk; CW corrects for it [Clark and West \(2007\)](#)
- Power: first-order autocorrelation of d_t -0.01 ; a gain of 0.2 standard deviations needs about 196 forecasts for 80% power, more than the 128 available
- Interpretation: “the Meese–Rogoff result holds for the leu over 2016–2026: the AR(1) does not beat the random walk, and our sample could only have detected a large gain”

B2: the term spread and growth [Proposed]

- **Question:** a team replicates [Estrella and Hardouvelis \(1991\)](#) and reports “the spread predicts US growth, $t = 4.7$ ”. Defend or correct it
- **Inputs:** FRED real GDP, 10-year and 3-month yields (quarterly averages), 1962–2025; $y_t = (400/4) \ln(\text{GDP}_{t+4}/\text{GDP}_t)$ on the spread; model: B1
- **Tasks**
 1. Estimate the slope with classical, Hansen–Hodrick-type (3 lags) and Newey–West standard errors.
 2. Defence question: why must the errors be at least MA(3)? Report the residual autocorrelation.
 3. Defence question: is the relation stable? Estimate it before 1989 and from 1989.
 4. Interpretation: rewrite the team’s sentence so that it matches the evidence.
- **Report:** a slope, three standard errors and t -statistics, one autocorrelation, two subsample slopes, one sentence
- **Notebook:** section B2

B3: a leakage audit [Solved]

- **Question:** a team selects the 5 best of 100 candidate predictors by their correlation with the target over the whole sample, then reports a positive out-of-sample R^2 on the last 60 periods. Is the gain real?
- **Inputs:** a simulation with pure-noise predictors and target, $T = 240$, 1 000 replications; model: B1
- **Tasks**
 1. Run the team's protocol and the honest protocol (selection on the training sample only).
 2. Report the mean out-of-sample R^2 and the share of positive values for both.
 3. Defence question: which line of code leaks, and how is it fixed?
 4. Interpretation: what would you require before believing a reported R^2 of 10%?
- **Report:** four numbers, one line of code, one sentence
- **Notebook:** section B3

B3: solution [Solved]

- ▣ Team protocol: mean $R^2 = 5.9\%$, positive in 81% of replications (90th percentile 14.4%)
- ▣ Honest protocol: mean $R^2 = -13.3\%$, positive in 8%: the predictors are noise
- ▣ The leak: the correlations are computed on `rows = slice(0, T)`; the fix is `slice(0, n_train)`, and in a rolling design the selection is repeated at every origin
- ▣ Interpretation: a R^2 of 10% is within the range that the leak alone produces; ask for the selection code, a placebo with shuffled targets and a test period never touched

B4: a rule search on the BET [Proposed]

- **Question:** a team reports “the 5-day moving-average rule beats buy-and-hold on the BET, $p = 0.006$ ”. It tried 20 rules. Defend or correct it
- **Inputs:** BET daily log returns, 2001–2012 (2993 days); rules “long if the close is above its n -day average”, $n = 5, 10, \dots, 100$; HAC t of the mean excess return; model: B3
- **Tasks**
 1. Compute the t -statistic of every rule and the naive p -value of the best.
 2. Compute the max- t p -value with the stationary bootstrap (mean block 10 days, 999 resamples).
 3. Defence question: why does the bootstrap resample the 20 series jointly, in blocks?
 4. Interpretation: what would you add before calling it a profitable rule?
- **Report:** 20 statistics, two p -values, one critical value, one sentence
- **Notebook:** section B4

C1: the pre-registration of your project [Proposed]

- **Question:** is your planned evaluation able to answer your question?
- **Inputs:** your team's landmark paper and data; the power formula of today's formula sheet; model: B1
- **Tasks**
 1. Write the seven sections of the pre-registration in at most one page: question and hypotheses, paper to replicate, data, models, evaluation, robustness, team.
 2. Write the rule that decides "replicated", "partly" or "not replicated".
 3. Compute the evaluation length needed for 80% power at the gain reported by the paper, with ρ estimated on your training data.
 4. Interpretation: if the needed length exceeds your data, how do you change the question?
- **Report:** one page, one rule, one number, one sentence
- **Notebook:** section C1

C2: audit an AI answer [Proposed]

- ▣ A student asked an AI assistant how to prepare the replication project and its defence. The answer:
- ▣ (a) Use the Diebold-Mariano test to compare your AR(1) with the random walk; it is valid for nested models.
- ▣ (b) With 19 donors, a placebo p-value of 0.01 is achievable if the effect is large.
- ▣ (c) If your result is not significant, the random walk is the true model.
- ▣ (d) For an AR(1) with $\phi = 0.6$, i.i.d. standard errors of the mean are too small by a factor of 2.
- ▣ (e) Standardise the predictors once on the full sample before the walk-forward loop: it saves time and changes nothing.
- ▣ (f) Holm rejects more hypotheses than Bonferroni and controls the same family-wise error rate.
- ▣ (g) You may use the assistant during the oral defence if you declare it in AI_USE.md.
- ▣ **Tasks**
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
 - ▶ 2. Report: a list of seven verdicts with one line of justification each.

Key takeaways

- ▣ Every number at the defence comes with its standard error, and the standard error with its assumption
- ▣ Nested models need Clark–West; a non-rejection needs a power calculation
- ▣ Overlapping targets and persistence enlarge standard errors; stability over subsamples is part of the evidence
- ▣ Selection on the whole sample creates out-of-sample gains from noise; a search needs a search-adjusted p-value
- ▣ Write the pre-registration and the verdict rule before the first estimate

After the seminar

- Lecture 15 develops each topic of today
 - ▶ the course map, one recap per chapter, the research workflow, the replication scoreboard
 - ▶ the failure modes, the oral defence and the AI policy
- Try the [Proposed] tasks in the notebook
- C1 is the first draft of your team's proposal and pre-registration
- Reading: [Clemens \(2017\)](#); [Nosek et al. \(2018\)](#); [Diebold \(2015\)](#)
- **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

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